•[1]:

```python
marks = [45, 67, 89, 32, 76]

marks.append(55)
marks.insert(2, 70)
marks.remove(32)
marks.sort()

print("Updated Marks List:", marks)
```

```
Updated Marks List: [45, 55, 67, 70, 76, 89]
```

[2]:

```python
cities = ("Bengaluru", "Delhi", "Mumbai", "Chennai", "Kolkata", "Hyderabad")

print("First city:", cities[0])
print("Last city:", cities[-1])
print("Total elements:", len(cities))
print("Is Delhi present?", "Delhi" in cities)
```

```
First city: Bengaluru
Last city: Hyderabad
Total elements: 6
Is Delhi present? True
```

Is Delhi present: True

[3]:
```python
A = {1,2,3,4,5}
B = {4,5,6,7}

print("Union:", A | B)
print("Intersection:", A & B)
print("Difference (A-B):", A - B)
print("Symmetric Difference:", A ^ B)
```

```
Union: {1, 2, 3, 4, 5, 6, 7}
Intersection: {4, 5}
Difference (A-B): {1, 2, 3}
Symmetric Difference: {1, 2, 3, 6, 7}
```

```
[4]: student = {
        "name": "Arjun",
        "roll": 101,
        "branch": "CSE",
        "marks": 85
    }

    student["marks"] = 92   # Update marks

    print("Keys:", student.keys())
    print("Values:", student.values())
    student = {
        "name": "Arjun",
        "roll": 101,
        "branch": "CSE",
        "marks": 85
    }

    student["marks"] = 92   # Update marks

    print("Keys:", student.keys())
    print("Values:", student.values())
```

```
Keys: dict_keys(['name', 'roll', 'branch', 'marks'])
Values: dict_values(['Arjun', 101, 'CSE', 92])
Keys: dict_keys(['name', 'roll', 'branch', 'marks'])
Values: dict_values(['Arjun', 101, 'CSE', 92])
```

[5]:

```python
items = ["Book", "Pen", "Bag", "Shoes"]
prices = [200, 20, 500, 1200]

total = sum(prices)
highest = max(prices)
lowest = min(prices)

print("Total Bill:", total)
print("Highest Price:", highest)
print("Lowest Price:", lowest)
```

```
Total Bill: 1920
Highest Price: 1200
Lowest Price: 20
```

[6]:

```
def add(a,b): return a+b
def sub(a,b): return a-b
def mul(a,b): return a*b
def div(a,b): return a/b

x = int(input("Enter first number: "))
y = int(input("Enter second number: "))

print("Addition:", add(x,y))
print("Subtraction:", sub(x,y))
print("Multiplication:", mul(x,y))
print("Division:", div(x,y))
```

```
Enter first number:  3
Enter second number:  4
Addition: 7
Subtraction: -1
Multiplication: 12
Division: 0.75
```

[7]:

```
def check_even_odd(n):
    return "Even" if n%2==0 else "Odd"

print(check_even_odd(7))
```

```
Odd
```

[8]:

```python
def largest(a,b,c):
    return max(a,b,c)

print("Largest:", largest(10,25,15))
```

```
Largest: 25
```

[9]:

```python
def largest(a,b,c):
    return max(a,b,c)

print("Largest:", largest(10,25,15))
```

```
Largest: 25
```

[10]:

```python
def is_prime(n):
    if n < 2: return False
    for i in range(2,int(n**0.5)+1):
        if n % i == 0:
            return False
    return True

print(is_prime(17))
```

```
True
```

[11]:
```
students = []

def add_student(name): students.append(name)
def remove_student(name): students.remove(name)
def display_students(): print(students)

add_student("Arjun")
add_student("Malika")
remove_student("Arjun")
display_students()
```

```
['Malika']
```

[12]:
```
employees = {}

def add_employee(name,salary): employees[name]=salary
def update_salary(name,salary): employees[name]=salary
def search_employee(name): return employees.get(name,"Not Found")

add_employee("Rahul",50000)
update_salary("Rahul",55000)
print(search_employee("Rahul"))
```

```
55000
```

[16]:

```python
nums = [int(input("Enter number: ")) for _ in range(10)]
unique = set(nums)
print("Unique count:", len(unique))
```

```
Enter number:  7
Enter number:  77
Enter number:  7
Enter number:  77
Enter number:  7
Enter number:  7
Enter number:  7
Enter number:  7
Enter number:  7
Enter number:  7
Unique count: 2
```

[17]:

```python
ages = (18, 25, 16, 40, 12)

def check_voting(age):
    return age >= 18

eligible = [age for age in ages if check_voting(age)]
print("Eligible ages:", eligible)
```

```
Eligible ages: [18, 25, 40]
```

[18]:

```python
sentence = input("Enter sentence: ")
words = sentence.split()
freq = {}

for w in words:
    freq[w] = freq.get(w,0) + 1

print(freq)
```

```
Enter sentence:  python is programming language
{'python': 1, 'is': 1, 'programming': 1, 'language': 1}
```

[19]:

```python
books = []

def add_book(name): books.append(name)
def remove_book(name): books.remove(name)
def search_book(name): return name in books
def display_books(): print(books)

add_book("Python Basics")
add_book("Data Structures")
remove_book("Python Basics")
display_books()
```

```
['Data Structures']
```

[20]:

```
def analyze(marks):
    avg = sum(marks)/len(marks)
    high = max(marks)
    low = min(marks)
    passed = [m for m in marks if m>40]
    return avg, high, low, passed

marks = [45,67,89,32,76]
print(analyze(marks))
```

```
(61.8, 89, 32, [45, 67, 89, 76])
```

[21]:

```
def check_password(pwd):
    return (any(ch.isdigit() for ch in pwd) and
            any(ch.isupper() for ch in pwd) and
            any(ch.islower() for ch in pwd) and
            len(pwd) >= 8)

print(check_password("StrongPass1"))
```

```
True
```

[22]:

```python
contacts = {}

def add_contact(name,number): contacts[name]=number
def search_contact(name): return contacts.get(name,"Not Found")
def delete_contact(name): contacts.pop(name,None)
def display_contacts(): print(contacts)

add_contact("Arjun","9876543210")
print(search_contact("Arjun"))
delete_contact("Arjun")
display_contacts()
```

```
9876543210
{}
```

[23]:

```python
balance = 0

def deposit(amount):
    global balance
    balance += amount

def withdraw(amount):
    global balance
    if amount <= balance:
        balance -= amount
    else:
        print("Insufficient funds")

def check_balance():
    print("Balance:", balance)

deposit(1000)
withdraw(500)
check_balance()
```

```
Balance: 500
```

[24]:

```python
lst = [10,20,30,40,50]

# Find max without built-in
max_val = lst[0]
for i in lst:
    if i > max_val:
        max_val = i

# Find min without built-in
min_val = lst[0]
for i in lst:
    if i < min_val:
        min_val = i

# Reverse without built-in
rev = []
for i in range(len(lst)-1,-1,-1):
    rev.append(lst[i])

print("Max:", max_val)
print("Min:", min_val)
print("Reversed:", rev)
```

```
Max: 50
Min: 10
Reversed: [50, 40, 30, 20, 10]
```